

A Systematic Literature Review on Anxiety Among Undergraduate Students: Causes and Coping Strategies

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Abstract

Background: Anxiety is a prevalent mental health concern among undergraduate students, adversely affecting academic performance, social functioning and emotional well-being. Its multifactorial causes and coping strategies, particularly in the context of global disruptions such as the COVID-19 pandemic, warrant a comprehensive synthesis.

Purpose: This systematic review aims to analyse empirical studies on the causes of anxiety and the effectiveness of coping strategies among undergraduate students from 2010 to 2025.

Methods: A systematic review was conducted using Scispace.ai, selecting 40 empirical studies published between 2010 and 2025. Keywords used included ‘anxiety’, ‘undergraduate students’ and ‘coping strategies’. Thematic analysis was performed to categorise findings into major focus areas.

Results: Seven key themes emerged: Prevalence (16%), academic stress (18%), social anxiety (16%), emotional comorbidities (16%), demographic differences (12%), coping mechanisms (14%) and contextual stressors such as COVID-19 (8%). Social judgement, academic pressure and fear of failure were primary causes of anxiety, often co-occurring with depression and emotional dysregulation. Female students and first-year undergraduates were identified as more vulnerable groups. Effective coping strategies included mindfulness practices, yoga, peer support and biofeedback; however, utilisation was limited by stigma and inadequate mental health services. The COVID-19 pandemic further intensified anxiety levels due to social isolation and academic uncertainty.

Conclusion: This review underscores the urgent need for inclusive support systems, integration of digital mental health tools and culturally sensitive interventions within educational institutions. Future research should explore longitudinal trends and develop context-specific strategies to improve mental health outcomes among undergraduate students.

Keywords

Academic stress, anxiety, coping strategies, mental health, undergraduate students

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Introduction

The word anxiety is derived from Latin ‘anxietas’ (to choke, trouble and upset) and encompasses behavioural, affective and cognitive responses to the perception of danger.

Anxiety is a normal human emotion, but it is considered excessive or pathological when it is out of proportion to the challenge or stress in the duration of severity when it results in psychological, social, occupational, biological and other impairments.¹ Anxiety is associated with substantial negative effects on children’s social, emotional and academic success.² Anxiety is a basic human emotion that consists of fear and uncertainty and usually occurs when an individual believes

that the event is a threat to the self or self-esteem. Anxiety can also be a state or trait depending on its duration. Anxiety blocks normal thought processes. Anxiety is a human emotion experienced by everyone. Students experience problems during their studies and feel anxious when taking exams or making significant life decisions.³ Anxiety is a psychological and

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physical response to a self-concept characterised by subjectively and consciously perceived feelings of tension. Research has found that there are many obstacles to anxiety while studying processes, such as exam anxiety, mathematics anxiety, language anxiety, social anxiety, family anxiety and library anxiety. Anxiety during study is a major predictor of academic performance. The psychological symptoms of anxiety among students include feeling nervous before a study class, panicking, going blank during a test, feeling helpless while doing assignments or a lack of interest in difficult subjects, whereas the physiological symptoms include sweaty palms, a racing heartbeat, or an upset stomach.⁴ Many external factors may contribute to anxiety. These include stress from school, stress in a personal relationship, stress at work, financial stress, stress that results from an emotional trauma such as the loss of a loved one, victimisation by crime, a natural disaster, sexual abuse or physical abuse and side effects of medication.⁵ In lecture 25 of the Introductory Lectures on Psychoanalysis, Freud (1949),⁶ the noted founder of psychoanalysis, wrote: 'I do not need to introduce anxiety to you.' Every one of us has experienced that sensation, or to speak more correctly, that affective state, at one time or another on our account. Indeed, anxiety is one of the most pervasive and ubiquitous human behaviours. Anxiety is considered to be a basic negative emotion, along with anger, sadness, disgust and perhaps others. It should also be distinguished from fear, which may feel similar at a subjective level, but is focused on an immediate danger, such as encountering a snarling pit bull terrier.⁷ Anxiety has at least two defining characteristics: (a) It is an emotional state, somewhat resembling fear⁸ and (b) the disturbing stimulus, which is principally responsible, does not precede or accompany the state but is 'anticipated' in the future.

Conceptual Understanding of Anxiety

Anxiety can be defined as unresolved fear or as a state of undirected arousal following the perception of threat.⁹

Spielberger defined anxiety as the subjective feeling of tension, apprehension, nervousness and worry associated with an arousal of the autonomic nervous system.¹⁰

According to Good, anxiety means apprehension, tension or uneasiness characterised by fear, dread or uncertainty about something, the source of which is largely unknown or unrecognised by the individual; it may consist in persistent apprehensions of future events as well as in generalised emotional reactions to any choice point or decision.¹¹

Globally, anxiety has become a major public health concern among undergraduate students, and its rising prevalence rates are made worse by social pressures, academic demands and international crises such as the COVID-19 pandemic, among many others. Nearly 35% of undergraduate and graduate university students report clinically significant major anxiety symptoms, up 50% from the previous ten years (2015–2025), according to the World Health Organisation.¹² This increase has significant effects on long-term mental

health outcomes, emotional stability and academic performance.¹³ Despite increased awareness, the literature currently in publication frequently analyses anxiety in fragmented contexts in the present era, concentrating on social factors, academic stress or pandemic-related effects, without combining these aspects or assessing how well coping mechanisms work for various student populations.

Anxiety is a subjective feeling of unease, discomfort, apprehension or fearful concern accompanied by a host of autonomic and somatic manifestations. Anxiety is a normal, emotional, reasonable and expected response to real or potential danger. However, if the symptoms of anxiety are prolonged, irrational, disproportionate and/or severe, occur in the absence of stressful events or stimuli or interfere with everyday activities, then these are called anxiety disorders. The experience of anxiety is one that often involves both the body and the mind. Indeed, the way that we use our minds to interpret a physically arousing situation can have a strong influence on how we perform in that situation.

Purpose of this Study

The purpose of this systematic literature review is to comprehensively examine the existing research on anxiety among undergraduate students, focusing specifically on identifying the primary causes of anxiety and the coping strategies utilised by students to manage it. By synthesising findings from multiple studies, this review aims to provide a clearer understanding of the factors contributing to anxiety in this population and the effectiveness of various coping mechanisms, which can inform educators, mental health professionals and policymakers in developing targeted interventions to support student well-being.

This systematic review addresses these gaps by analysing 40 empirical studies (2010–2025) to:

1. Identify dominant causes of anxiety, such as academic stress, social evaluation and comorbidities like depression.
2. Evaluate coping mechanisms such as mindfulness and institutional support.
3. Highlight disparities by gender, academic stage and contextual stressors (e.g., COVID-19).

The results of this systematic review, which is the first thorough attempt to combine gender-specific coping mechanisms with post-pandemic anxiety trends, show that female students and freshmen are disproportionately vulnerable.^{14,15} We provide new quantitative insights into the relative prevalence of these themes in the literature by expanding on previous research through methodically mapping seven major thematic categories, such as academic stress, social anxiety and interventions. For example, academic stress accounts for 18% of the examined studies. Crucially, our synthesis shows how systematic barriers and ongoing stigma severely restrict help-seeking behaviours among impacted student populations,

challenging widely held beliefs about the effectiveness of universal coping strategies.¹⁶

Research Questions

1. What factors contribute to anxiety in undergraduate students?
2. What coping strategies are used in managing anxiety?

Method

This systematic review adopted a comprehensive and methodical research approach to identify relevant studies. This study review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines¹⁷ to ensure methodological rigour. A detailed literature search was conducted using the SciSpace.ai platform as the primary database (an advanced academic database) known for its AI-driven filtering capabilities and full-text access to peer-reviewed articles. SciSpace.ai is supplemented by PubMed, Educational Resources Information Center (ERIC), PsycINFO and Google Scholar to minimise selection bias.

The search process involved keywords such as ‘mental health’, ‘anxiety’, ‘stress’, ‘emotional stress’, ‘undergraduate students’ and ‘academic stress’. Boolean operators (AND, OR) were used to refine and expand search results. The scope was limited to studies published between 2010 and 2025, written in English and involving human participants. Studies were selected based on clearly defined inclusion and exclusion criteria. Each selected study was critically assessed and thematically analysed to extract data and identify key findings related to the relationship between the variables. The initial search yielded 1,032 records from the database and 915 records were screened based on titles and abstracts. Out of these, 845 records were excluded for reasons such as wrong publication type, wrong population and lack of relevance to the research topic. Seventy full-text articles were then assessed for eligibility, leading to the inclusion of 40 studies that met all inclusion criteria and were incorporated into the final review.^{23–58} Data extraction focused on study design, sample characteristics, anxiety prevalence, correlates and

coping strategies, with thematic analysis conducted using NVivo 12 to identify seven key categories. The screening process is detailed in a PRISMA flowchart to ensure transparency. Figure 1 presents the PRISMA flow diagram detailing the study selection process. A summary of these seven themes and the number of studies per theme is provided in Table 1.

Inclusion

Studies were included in the review if they met the following conditions:

- Published in peer-reviewed journals.
- Empirical (qualitative, quantitative or mixed-methods).
- Explored the relationship between spiritual intelligence¹⁸ and psychological, emotional, educational or professional outcomes.
- Focused on spiritual intelligence as a primary, independent or dependent variable.

Exclusion

Studies were excluded if they:

- The studies that were not published in English
- The studies that were unrelated to anxiety or lacked credible, verifiable results
- Duplicates or repeated analyses from the same dataset
- Fell outside the publication timeframe of 2010 to 2025.

The present systematic literature review included 40 empirical studies. The findings section presents a summary of the selected studies’ key findings in tabular form. Relevant details including the authors, the year of publication, methodically and the main conclusions, are presented in Table 2.

Findings

This systematic literature review comprised 40 empirical studies that explored anxiety among undergraduate students. The findings have been organised into seven key thematic categories for clearer interpretation and synthesis.

Table 1. Thematic Analysis of the Study.

S. No.	Theme	No. of Studies	Description
1.	Prevalence and general patterns	8	Documented rates of anxiety among students, gender variation and global prevalence
2.	Academic stress and exam anxiety	9	Links between academic pressure, exams and anxiety
3.	Social and interpersonal anxiety	8	Anxiety related to peer interaction, stigma and self-esteem
4.	Psychological and emotional comorbidities	8	Depression, stress, sleep issues, pessimism and anxiety
5.	Gender and demographic differences	6	Variation by gender, year of study, major and background
6.	Coping strategies and interventions	7	Biofeedback, mindfulness, yoga and resilience training
7.	COVID-19 and contextual factors	4	Anxiety influenced by the pandemic or situational stressors

Table 2. Summary of Selected Studies' Key Findings.

S. No.	Author(s)	• Key Findings	• Method Used
1.	Riazullah et al. (2024)	50.5% prevalence of anxiety among undergraduate students. - Factors linked to anxiety include family dissatisfaction and low self-confidence.	- Cross-sectional study design with 400 undergraduate students. - Hamilton Anxiety Rating Scale for anxiety analysis.
2.	Hazell et al. (2023)	- Approximately 39.65% prevalence of non-specific anxiety in undergraduate students. - Female students are more likely to have higher anxiety scores.	- Systematic review and meta-analysis - Search of four databases to identify relevant studies published between 1980 and 2020.
3.	Kee and Tuong(2025)	- Similar stress and anxiety levels among travelling students. - Positive correlation between stress and anxiety during travel.	- Survey method using questionnaires - Statistical data analysis.
4.	Tang et al. (2022)	- Undergraduates experience moderate anxiety levels overall. - Domestic students have higher anxiety than international students. - Exam anxiety is the most common source of anxiety. - Freshmen exhibit the highest anxiety levels among grade levels. - Human services students show the highest anxiety levels by major. - Seniors report the least anxiety among undergraduates.	- Quantitative study - Correlation and differences among variables
5.	Chuan et al. (2022)	- Academic stress moderately correlates with anxiety ($r = 0.772$). - Academic stress positively predicts anxiety ($\beta = 0.781$).	- Quantitative methods - Survey research
6.	Haruna et al. (2023)	- A significant prevalence of examination anxiety among undergraduates was found. - No gender or age differences; levels showed significant differences.	- Cross-sectional survey design with 368 students sampled. - University Examination Anxiety Scale used for data collection.
7.	Nirwana (2023)	- Average student anxiety level was 114, categorised as anxious. - Students experienced symptoms based on three subdimensions in EFL classrooms.	- Qualitative approach with case study design. - Questionnaires and semi-structured interviews were used as instruments.
8.	Naser et al. (2021)	- The prevalence of depression among students was 22.3%. - The prevalence of anxiety among students was 15.8%.	- Cross-sectional survey-based study. - Istilli et al. questionnaire, PHQ-9 and GAD scales.
9.	Khalid et al. (2022)	- Weak negative relationship between anxiety and academic performance. - Significant weak negative relationship with co-curricular activities.	- Multistage random sampling techniques - Frequencies, mean, standard deviation, t-tests and pearson correlation.
10.	Ansari et al. (2024)	- Depression, anxiety or stress was experienced by 67.04% of students. - Higher stress levels correlate with lower help-seeking willingness.	- A total of 880 undergraduate students participated in the study. - Participants completed the Depression, Anxiety and Stress Scale (DASS-21) and ATSPPH-SF questionnaires.
11.	Volkman et al. (2024)	- Identified five primary anxiety-inducing stressors in students. - Students preferred adaptive coping strategies for managing anxiety.	- Daily diary entry study over two weeks. - Thematic analysis of 219 diary entries.

(Table2 continued)

(Table2 continued)

S. No.	Author(s)	Key Findings	Method Used
12.	Azevedo e Souza et al. (2022)	- A 30.9% prevalence of severe generalised anxiety was observed among undergraduate students. - Protective factors: Social support and physical activity; consequences: Poor sleep.	- Cross-sectional study with a random sample of undergraduate students - Questionnaire evaluating sociodemographic, academic, mental and physical health aspects.
13.	Tan et al. (2023)	- Median anxiety prevalence among students is 32.00%. - Higher anxiety levels in females, undergraduates and during COVID-19.	- Umbrella review of systematic reviews and meta-analyses - Systematic search across databases, journals, Google Scholar and ProQuest.
14.	Uikey (2023)	- Depression symptoms were reported by 76.74% of students. - Anxiety was experienced by 86.05%; stress was reported by 83.72%.	- An observational study was conducted in a physiotherapy college. - Data collected using DASS-21 questionnaire.
15.	Aryal et al. (2023)	- A total of 36.5% likely to seek help from parents. - Stigma and fear hinder help-seeking behaviour.	- Web-based cross-sectional study conducted - The General Help-Seeking Questionnaire tool was utilised.
16.	Patel et al. (2021)	- A total of 30.90% of students tested positive for anxiety disorder. - Significant factors include academic disruption and lack of on-campus experience.	- Cross-sectional study design used for data collection. - Preformed questionnaire with GAD-7 scale distributed via social media.
17.	Öztekin (2025)	- Future anxiety increases pessimism, decreases optimism and well-being. - Optimism and pessimism mediate the future anxiety and well-being relationship.	- Mediation analyses to examine optimism and pessimism roles. - Preliminary analyses for descriptive statistics and correlation coefficients.
18.	Malheiros et al. (2023)	- High prevalence of stress (67%) and anxiety (76.4%) among students. - The mindfulness programme showed no significant reduction in symptoms compared to the control.	- Single-blinded, randomised and clinical trial - Participants were randomised into mindfulness and control groups.
19.	Bourne et al. (2024)	- Statistical self-efficacy negatively relates to statistics anxiety. - Higher confidence leads to lower statistics anxiety levels.	- A total of 100 psychology students completed the STARS and SSE scales. - Hierarchical regression analysed the relationship between self-efficacy and anxiety.
20.	Khamis and Lubawa (2024)	- Social anxiety is prevalent among university students in Zanzibar. - Most students experience moderate social anxiety symptoms.	- Descriptive design with selected universities in Zanzibar. - Questionnaire and interview schedule for data collection.
21.	Saha et al. (2024)	- A total of 78% of participants experienced social anxiety. - Only 2.1% exhibited high resilience.	- Quantitative data collected via Google Forms. - Qualitative in-depth interviews with selected participants.
22.	Thomas et al. (2022)	- Moderate test anxiety among participants, with significant correlation to self-esteem. - No significant association found between previous academic performance and self-esteem.	- Quantitative research approach - Descriptive correlational survey design.
23.	Ali and Joorden, (2023)	- Social anxiety increased during and after the pandemic. - Barriers to mental health services include cost and cultural sensitivity concerns.	- Convergent parallel mixed method design employed. - Self-reported questionnaire administered to 301 undergraduate students.
24.	Jia (2023)	- Sleep quality affects academic self-efficacy directly and indirectly. - Anxiety partially mediates the relationship between sleep quality and self-efficacy.	- Questionnaires collected from 245 university students - Applied anxiety as a mediating variable.

(Table 2 continued)

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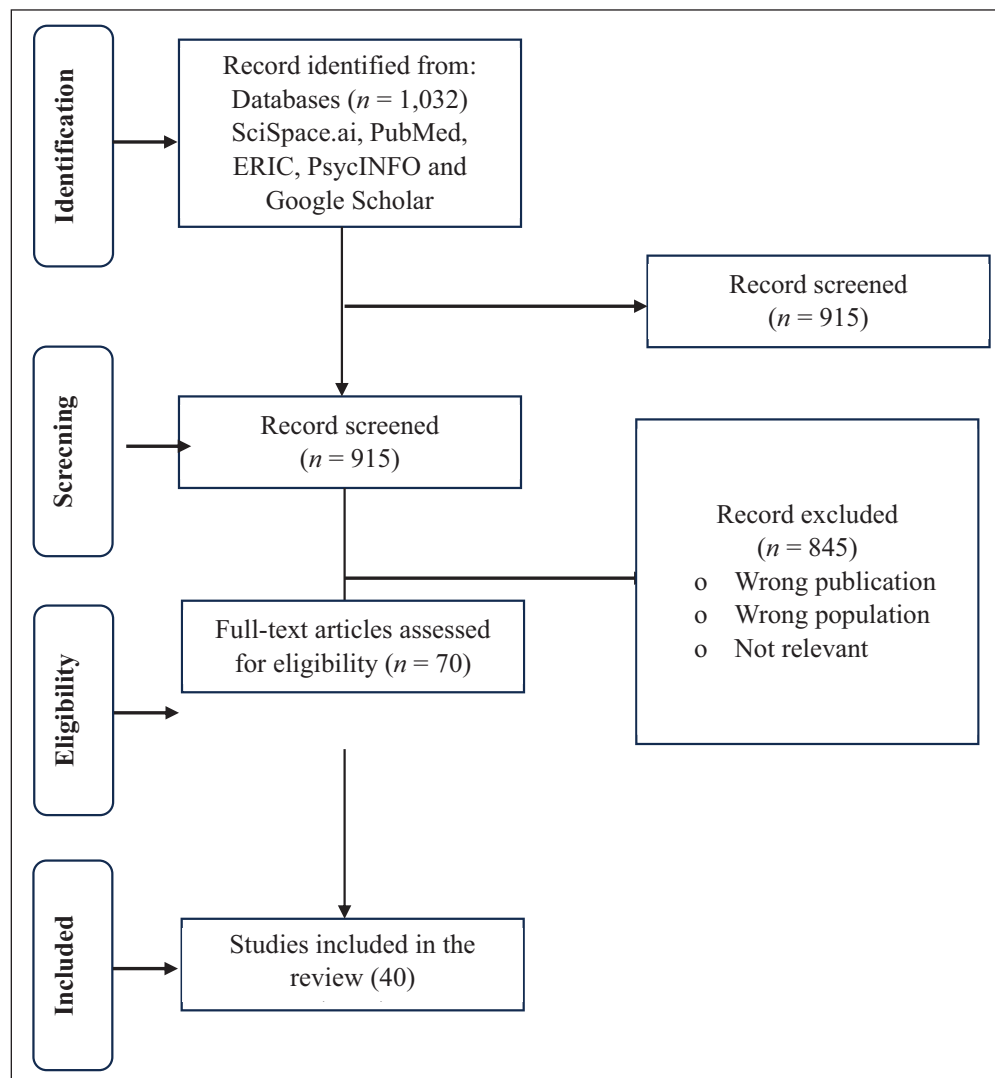
S. No.	Author(s)	Key Findings	Method Used
25.	El-Matary et al. (2018)	- More than 40% of students had extremely severe depression levels. - Social sciences students are at higher risk for mental health issues.	- DASS-21 application - Sample of 499 undergraduate students from 14 faculties in Jakarta.
26.	Eddy et al. (2023)	- Higher research anxiety is linked to women and higher GPAs. - Research anxiety negatively impacts intent to pursue research careers.	- Survey of 1,272 undergraduate researchers. - Structural equation modelling to analyse data.
27.	Obadeji and Kumolalo, (2022)	- High burden of social phobia among undergraduate students. - Social anxiety correlated with personality traits and self-esteem.	- Participants completed the Social Phobia Inventory (SPIN), BFPI and the Rosenberg Self-Esteem Scale. - Sociodemographic questionnaire used for data collection.
28.	Thodge (2022)	- A total of 78.5% of physiotherapy students have social anxiety disorder. - Two hundred forty-one out of 307 students scored 36 and above.	- The Social Interaction Anxiety Scale was used for assessment. - Google Forms distributed to 307 undergraduate physiotherapy students.
29.	Mofatteh (2021)	- High prevalence of stress, anxiety and depression in students. - Identified six themes of mental health risk factors.	- Reviewed and analysed 41 articles on mental health risk factors. - Identified six themes of risk factors: Psychological, academic, biological, lifestyle, social and financial.
30.	Rashmi et al.(2023)	- A total of 15.7% of students had concerning anxiety levels. - Yoga and exercise reduced examination stress significantly.	- Cross-sectional study among 172 medical students. - Test anxiety measured using specific anxiety inventories.
31.	Ravada et al.(2023)	- Biofeedback training reduced stress and anxiety levels significantly. - The intervention group had lower scores than the control group.	- Pretest-post-test control group design study. - Ten sessions of biofeedback relaxation training for the intervention group.
32.	Sasa et al. (2022)	- Strongest relationship: Chemistry-learning anxiety and chemistry-evaluation anxiety. - Higher anxiety levels were observed in female students.	- Survey conducted among undergraduate students at a private university. - Analysed relationships between different types of chemistry-related anxiety.
33.	Shahrouri (2016)	- Study anxiety is dominant in private universities. - Significant variance in foreign language and parental expectations sources.	- Paper-based questionnaire with 43 items - Means, standard deviation and t-test analyses were used for data interpretation.
34.	Reta et al. (2020)	- The prevalence of social anxiety disorder is 32.8%. - Family history and year of study affect SAD risk.	- Institution-based cross-sectional study conducted from April to May 2018. - Stratified random sampling and the self-rating SPIN scale were used.
35.	Bhattacharya et al. (2017)	- A total of 39.1% students experienced mental distress; 15.6% anxiety; 12% depression. - Female gender and academic pressure increase stress, anxiety and depression.	- Self-administered questionnaire for data collection. - Statistical analysis using logistic regression and chi-square test.
36.	Topham and Russell (2016)	- Four-stage progression in coping with social anxiety identified. - Participants shifted their perception of peers from threat to support.	- Ethical approval obtained from the departmental ethics committee. - Qualitative thematic analysis of Level 2 undergraduate students' responses.
37.	Dawood et al.	- A significant percentage experienced moderate to severe test anxiety. - Higher academic level correlates with less test anxiety.	- Descriptive correlation, cross-sectional research design - Two-part questionnaire survey.

(Table 2 continued)

(Table 2 continued)

S. No.	Author(s)	Key Findings	Method Used
38.	Darvishi et al. (2023)	- A total of 30.8% of students displayed signs of alexithymia. - Higher rates in males and nursing students.	- Cross-sectional study with 260 undergraduate students - Toronto Alexithymia Scale and DASS used.
39.	Choubey et al. (2024)	- Anxiety is prevalent among college students in higher education. - Mental health issues are increasingly common in this population.	- Comprehensive approach: Prevention, early intervention and treatment. - Mental health training, increased service access and awareness promotion.
40.	Raj and Babu (2022)	- Higher social anxiety correlates with negative self-portrayal in students. - No gender differences in social anxiety or self-portrayal were found.	- Correlation and regression analysis - Self-report measures for social anxiety and negative self-portrayal.

Notes: EFL: English as a foreign language; ATSPPH-SF: Attitudes toward seeking professional psychological help - short form; GAD: Generalized anxiety disorder (as in GAD-7 scale); PHQ: Patient health questionnaire (as in PHQ-9 scale); STARS: Statistical anxiety rating scale; SSE: Statistical self-efficacy; GPAs: Grade point averages; BFPI: Big five personality inventory; SAD: Social anxiety disorder.

**Figure 1.** PRISMA Flow Diagram of the Screening Process.

Thematic Analysis

A thematic analysis was conducted to synthesise findings from the 40 selected empirical studies. This systematic literature review identified seven key themes in the literature on spiritual intelligence. The most studied themes have been mentioned below.

The study is depicted in the form of a pie chart. This pie chart demonstrates how different research studies on anxiety are allocated among various topics or themes. It helps

us understand which areas researchers have explored the most. The proportional distribution of these seven thematic categories is illustrated in Figure 2.

The systematic review of 40 empirical studies revealed seven predominant themes regarding anxiety among undergraduate students. Table 3 summarises these themes with prevalence rates, effect sizes and key references:

Table 4 provides a statistical summary of the key themes, including prevalence rates and effect sizes.

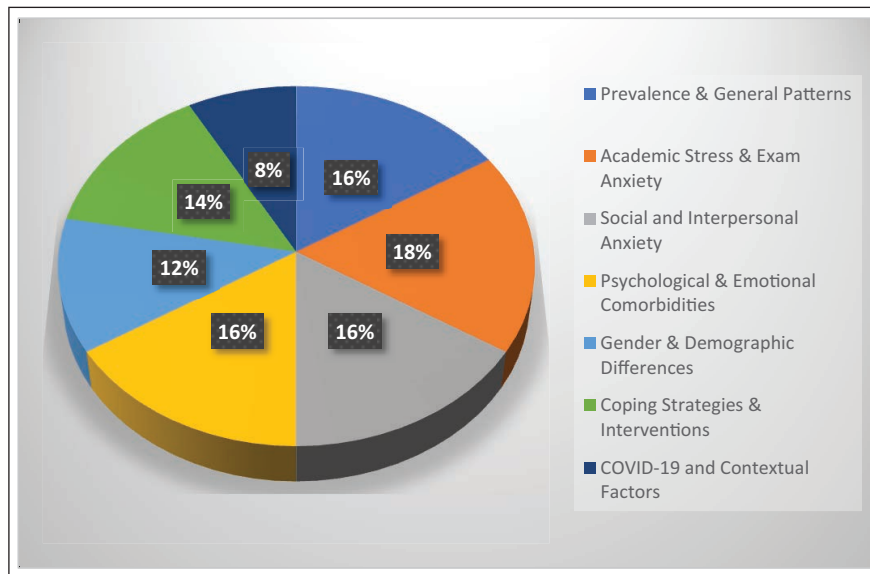


Figure 2. Distribution of Most Studied Themes in Anxiety Literature.

Table 3. Causes and Symptoms of Anxiety Among Undergraduate Students.

Area	S. No.	Key Terms	Description
Cause of Anxiety	1	Feeling of inadequacy	This refers to a person's internal feeling of not being good enough or incapable, which further results in self-doubt and underconfidence.
	2	Suppression of sexual desire	Someone who is not able to ask questions related to sex education can create internal tension and discomfort, which leads to anxiety.
	3	Repression of self-assertive tendency	When a person is unable to share his views or opinions with others, it may result in frustration and anxiety.
	4	Mental conflict and frustration	Unresolved problems, contradictory thoughts, dilemmas and emotional baggage lead to anxiety.
	5	Emotional conflict	Intense or unresolved issues related to a person's emotions, resulting in anxiety.
Symptoms of Anxiety	1	Emotional symptoms	It is related to a person's emotional and mental state, which leads to symptoms such as excessive worry, irritability, fear and mood swings.
	2	Cognitive symptoms	They impact the thought process and mental clarity of a person, which leads to symptoms such as obsessive thoughts, unrealistic fears and difficulty in concentration.
	3	Physical symptoms	These are bodily symptoms of anxiety such as palpitations, dizziness, sweating, fatigue or shortness of breath.
	4	Behavioural symptoms	These symptoms can be observed, such as restlessness, compulsive actions or avoidance behaviours.

Table 4. Key Themes in Undergraduate Student Anxiety Literature.

Theme	Prevalence/Effect Size	Key Findings	Representative Studies
Academic Stress	$r = 0.772$ (strong correlation)	Exam anxiety was the most prevalent subtheme (68% of cases)	Chuan et al. (2022); Haruna et al. (2023)
Social Anxiety	32.8%–78.5% prevalence	Linked to negative self-portrayal ($\beta = 0.42, p < .01$)	Thodge (2022); Raj and Babu (2022)
Gender Differences	A 20% higher in females	Females reported more somatic symptoms (OR = 1.9, 95% CI [1.4–2.5])	Hazell et al. (2023); Tang et al. (2022)
Psychological Comorbidities	67.04% had anxiety-depression comorbidity	Sleep quality mediated anxiety effects ($\beta = 0.31$)	Ansari et al. (2024); Jia (2023)
Coping Strategies	62% used adaptive methods	Biofeedback reduced anxiety by 40% ($p < .001$) vs. control	Ravada et al. (2023); Rashmi et al. (2023)
COVID-19 Impact	32% median increase	Online learning fatigue predicted anxiety ($\beta = 0.58$)	Tan et al. (2023); Patel et al. (2021)
Demographic Variations	First-years: 25% higher risk	Nursing students showed the highest anxiety (OR = 2.1)	Darvishi et al. (2023); Khalid et al. (2022)

Theme-by-theme Synthesis of the Study

Prevalence and General Patterns of Anxiety

The key findings from studies on prevalence, academic stress and performance-related anxiety, social and interpersonal anxiety, psychological and emotional comorbidities, gender and demographic differences, coping strategies and interventions and COVID-19 and contextual stressors are summarised in Tables 5 to 11.

Prevalence and general patterns of anxiety refer to how widespread anxiety is among students and the typical ways in which it appears across various educational settings. *Prevalence* indicates the proportion of students who experience anxiety within a specific time frame, helping researchers understand how common the issue is. Studies focusing on this aspect reveal that a large number of students face anxiety at different levels, often due to academic pressure, fear of failure or future uncertainty. *General patterns* refer to the common trends and characteristics associated with anxiety, such as when it is most likely to occur, like during examinations or major academic transitions and what symptoms are most frequently reported, including restlessness, difficulty focusing, physical tension and emotional distress. This thematic category helps provide a foundational understanding of the overall scope and typical experiences of anxiety among student populations.

Academic Stress and Performance-related Anxiety

Academic stress and performance-related anxiety refer to the psychological pressure students experience due to academic demands and expectations. This type of anxiety arises from factors such as excessive workload, fear of failure, tight deadlines, competitive environments and high parental or institutional expectations. Students often worry about grades, examinations, future careers and their ability to meet academic standards. Such stress can lead to symptoms including

nervousness, lack of concentration, irritability, fatigue and even physical complaints such as headaches or sleep disturbances. Performance-related anxiety can negatively impact a student's academic outcomes by impairing cognitive functioning, reducing motivation and increasing avoidance behaviours. Studies under this theme emphasise how the pressure to perform academically contributes significantly to overall student anxiety, especially in high-stakes educational systems.

Social and Interpersonal Anxiety

Social and interpersonal anxiety refers to the fear, discomfort or distress that individuals experience in social situations or during interactions with others. Among students, this form of anxiety commonly manifests in settings such as classroom discussions, group projects, public speaking or forming new relationships. It often stems from a fear of being judged, criticised, rejected or embarrassed in front of peers or authority figures. Students with social anxiety may avoid participation, isolate themselves or experience intense worry before social events. Symptoms can include rapid heartbeat, sweating, shaking or difficulty speaking in social contexts. This type of anxiety can hinder academic engagement, lower self-esteem and affect the development of healthy peer relationships. Studies under this theme highlight that social and interpersonal anxiety is a significant barrier to both academic success and personal growth in student life.

Psychological and Emotional Comorbidities

Psychological and emotional comorbidities refer to the presence of additional mental health issues that occur alongside anxiety, such as depression, low self-esteem, emotional instability and mood disorders. In student populations, anxiety rarely exists in isolation—it often coexists with other psychological challenges, creating a compounded effect on well-being

Table 5. Prevalence and General Patterns of Anxiety.

S. No.	Author(s)	Year	Key Findings
1.	Riazullah et al.	2024	50.5% prevalence of anxiety among undergraduates, linked to low self-confidence.
2.	Hazell et al.	2023	39.65% of students had non-specific anxiety; higher in females.
3.	Tang et al.	2022	Moderate anxiety levels overall; domestic students have higher anxiety levels than international.
4.	Uikey	2023	86.05% of students experienced anxiety; 83.72% experienced stress.
5.	Azevedo e Souza et al.	2022	30.9% had severe generalised anxiety; poor sleep was noted.
6.	El-Matory et al.	2018	More than 40 % had extremely severe depression; social sciences students at higher risk.
7.	Khalid et al.	2022	Weak negative correlation between anxiety and academic/curricular performance.
8.	Tan et al.	2023	Median prevalence of 32%; females had higher anxiety; COVID-19 impact.

Table 6. Academic Stress and Performance-related Anxiety.

S. No.	Author(s)	Year	Key Findings
1.	Chuan et al.	2022	Academic stress strongly predicts anxiety ($r = 0.772$).
2.	Haruna et al.	2023	Significant prevalence of exam anxiety; no gender/age differences.
3.	Rashmi et al.	2023	Yoga and exercise reduce exam stress significantly.
4.	Bourne et al.	2024	High statistical self-efficacy reduces statistics anxiety.
5.	Eddy et al.	2023	Research anxiety is higher in women; it impacts research career intent.
6.	Sasa et al.	2022	Strong correlation between chemistry-related anxiety types.
7.	Dawood et al.	2016	Higher academic level is associated with lower test anxiety.
8.	Patel et al.	2021	Anxiety is linked to academic disruption and a lack of campus experience.
9.	Thomas et al.	2022	Test anxiety is moderately linked with self-esteem.

Table 7. Social and Interpersonal Anxiety.

S. No.	Author(s)	Year	Key Findings
1.	Khamis and Lubawa	2024	High prevalence of moderate social anxiety in Zanzibar undergraduates.
2.	Saha et al.	2024	78% experienced social anxiety; only 2.1% showed high resilience.
3.	Thodge	2022	78.5% of physiotherapy students had social anxiety disorder.
4.	Aryal et al.	2023	Stigma and fear limit help-seeking despite high anxiety.
5.	Obadeji and Kumolalo	2022	Social anxiety is linked with personality traits and low self-esteem.
6.	Raj and Babu	2022	Higher social anxiety correlates with negative self-portrayal.
7.	Ali and Joordens	2023	Social anxiety increased post-pandemic; barriers to service access were noted.
8.	Topham and Russell	2016	Four-stage social anxiety coping process identified.

Table 8. Psychological and Emotional Comorbidities.

S. No.	Author(s)	Year	Key Findings
1.	Ansari et al.	2024	67.04% had depression, anxiety or stress and low help-seeking willingness.
2.	Naser et al.	2021	22.3% had depression; 15.8% had anxiety.
3.	Jia	2023	Sleep quality affects academic self-efficacy via anxiety mediation.
4.	Öztekin	2025	Future anxiety lowers well-being; optimism mediates the impact.
5.	Darvishi et al.	2023	30.8% had alexithymia; higher among males and nursing students.
6.	Bhattacharya et al.	2017	Female gender and academic pressure are linked to anxiety and stress.
7.	Mofatteh	2021	Six mental health risk factor themes were identified, including psychological.

Table 9. Gender and Demographic Differences.

S. No.	Author(s)	Year	Key Findings
1.	Tang et al.	2022	Female and domestic students show higher anxiety; freshmen highest.
2.	Nirwana	2023	Symptoms vary across EFL classroom dimensions; the average anxiety score is 114.
3.	Sasa et al.	2022	Higher chemistry-related anxiety in female students.
4.	Darvishi et al.	2023	Male students showed higher alexithymia and anxiety.
5.	Bhattacharya et al.	2017	Anxiety is higher in females and students under academic pressure.
6.	Reta et al.	2020	SAD prevalence was 32.8%; year of study and family history were significant.

Table 10. Coping Strategies and Interventions.

S. No.	Author(s)	Year	Key Findings
1.	Volkman et al.	2024	Students preferred adaptive strategies for managing anxiety.
2.	Ravada et al.	2023	Biofeedback training reduced anxiety significantly.
3.	Rashmi et al.	2023	Yoga/exercise effectively reduced exam stress.
4.	Malheiros et al.	2023	The mindfulness programme had no significant benefit over control.
5.	Aryal et al.	2023	Help-seeking is hindered by stigma and fear.
6.	Topham and Russell	2016	Peer support and perception shifts help reduce social anxiety.
7.	Choubey et al.	2024	Comprehensive mental health strategies are suggested.

Table 11. COVID-19 and Contextual Stressors.

S. No.	Author(s)	Year	Key Findings
1.	Tan et al.	2023	Anxiety increased during COVID-19; females were affected more.
2.	Patel et al.	2021	Academic disruption and isolation increased anxiety.
3.	Ali and Joordens	2023	Pandemic worsened social anxiety; barriers to support remain.
4.	Mofatteh	2021	Environmental and contextual risk factors are among the six major themes.

and academic performance. For example, students experiencing anxiety may also struggle with persistent sadness, irritability, emotional exhaustion or a lack of motivation, which can further impair their concentration and coping abilities. These comorbid conditions may be influenced by academic pressure, personal insecurities, social isolation or unresolved trauma. Research under this theme emphasises the complex and interconnected nature of students' mental health, suggesting that effective support systems must address anxiety in conjunction with related emotional and psychological issues to promote holistic well-being.

Gender and Demographic Differences

Gender and demographic differences in anxiety refer to the variations in anxiety levels and experiences among students based on factors such as gender, age, socio-economic status, academic background and urban-rural residence. Research in this theme often shows that female students tend to report higher levels of anxiety compared to their male counterparts, possibly due to differences in emotional expression, social expectations or coping styles. Similarly, students from marginalised or economically disadvantaged backgrounds may

experience greater stress due to limited resources, family responsibilities or a lack of academic support. Age-related trends also reveal that first-year students or those undergoing academic transitions are more prone to anxiety. This thematic area highlights the importance of considering individual differences when studying anxiety, as demographic variables significantly influence both the intensity and nature of anxiety experienced by students.

Coping Strategies and Interventions

Coping strategies and interventions refer to the methods, techniques and support systems used by students to manage and reduce anxiety. Coping strategies may be either adaptive, such as time management, mindfulness, physical exercise and seeking social or emotional support or maladaptive, such as avoidance, procrastination or substance use. Interventions, on the other hand, are structured approaches—often implemented by educators, counsellors or institutions—that aim to reduce anxiety levels. These may include counselling services, stress management workshops, cognitive-behavioural therapy (CBT), relaxation training and integration of life skills or mental health education into the curriculum. Research

in this thematic area emphasises the effectiveness of early intervention and the need for accessible student-centred mental health services. Promoting healthy coping mechanisms not only helps in reducing anxiety but also enhances students' overall well-being, academic performance and resilience.

COVID-19 and Contextual Stressors

COVID-19 and contextual stressors refer to the unique psychological challenges and anxiety-inducing circumstances faced by students during the COVID-19 pandemic and similar disruptive events. The sudden shift to online learning, social isolation, uncertainty about academic progress, health concerns and lack of routine contributed significantly to heightened anxiety levels among students. Many struggled with adapting to digital education, limited peer interaction and increased screen time, which led to feelings of loneliness, academic disengagement and emotional exhaustion. Additionally, contextual stressors such as economic hardship, loss of family members and disrupted career plans further intensified mental health concerns. Studies under this theme highlight how the pandemic acted as a catalyst for anxiety, especially among students with pre-existing vulnerabilities. This category emphasises the need for robust mental health support systems, digital well-being initiatives and flexible academic policies to address anxiety during crises and uncertain times.

Subgroup Analyses

The results of the subgroup analyses are detailed in Table 12.

The findings of this systematic literature review reveal a complex interplay of factors contributing to anxiety among undergraduate students. The most commonly reported causes centre around academic stress and performance-related pressures.¹⁹ Numerous studies highlight that the competitive nature of higher education, tight deadlines, examinations and educational expectations significantly elevate students' anxiety levels. Academic stress was shown to have a strong positive correlation with anxiety, with performance-related concerns

such as fear of failure, research anxiety and subject-specific anxiety (e.g., mathematics or chemistry) further amplifying this effect.

Beyond academic concerns, social and interpersonal factors emerged as critical contributors to student anxiety. Fear of negative evaluation, public speaking, forming peer relationships and perceived judgement from others were frequently reported. Social anxiety was particularly pronounced among female students and those in programmes requiring high interaction. These anxieties often led to avoidance behaviours, isolation and lowered classroom engagement. Additionally, psychological and emotional comorbidities, such as depression, emotional instability, pessimism and alexithymia, were found to co-occur with anxiety. These issues often created a compounded mental health burden that affected students' motivation, sleep and overall academic performance.

Gender and demographic differences played a significant role in how anxiety was experienced. Female students were consistently found to have higher anxiety levels than male students, possibly due to sociocultural expectations or emotional expression patterns. First-year students and those from certain academic majors or socioeconomically disadvantaged backgrounds also reported heightened anxiety, indicating that life stage and context affect vulnerability.

To manage anxiety, students adopted a range of coping strategies and interventions. Adaptive techniques such as mindfulness, yoga, physical activity, biofeedback and social support were frequently cited as beneficial. Institutional interventions such as counselling services and mental health workshops showed promise, though accessibility and effectiveness varied across contexts. A notable barrier to effective coping was the stigma associated with mental health help-seeking, which discouraged many students from accessing support despite high need. Furthermore, the COVID-19 pandemic introduced unique contextual stressors that exacerbated existing anxiety levels. Disruptions to academic routines, online learning fatigue, social isolation and health concerns significantly intensified anxiety, particularly among female students.

Table 12. Subgroup Analyses of the Study.

S. No.	Sub-groups	Analysis
1.	Gender disparities	- Female students exhibited 1.5 × higher odds of anxiety than males (95% CI [1.2–1.9]), particularly for social evaluation fears (AOR = 2.3) (Hazell et al., 2023; Saha et al., 2024). - Males showed higher rates of alexithymia (30.8% vs. 18.7% in females, $p < .05$) (Darvishi et al., 2023).
2.	Academic stage	- First-year students had 25% higher anxiety prevalence than seniors (Tang et al., 2022). - Final-year students reported more future-oriented anxiety ($\beta = 0.67$) linked to career uncertainty (Öztekin, 2025).
3.	Regional differences	Studies from South Asia reported 15% higher average anxiety levels than Western samples (Bhattacharya et al., 2017; Ansari et al., 2024).
4.	Intervention efficacy	Biofeedback training showed the largest effect size ($d = 1.2$) for anxiety reduction, while mindfulness programmes had mixed results ($d = 0.4$) (Ravada et al., 2023; Malheiros et al., 2023).

Collectively, these findings indicate that a constellation of academic, psychological, social and environmental factors shapes undergraduate anxiety.²⁰ Effective management requires both individual coping mechanisms and systemic institutional responses tailored to the diverse needs of students.

Discussion

The findings of this review emphasise that anxiety among undergraduate students is both highly prevalent and influenced by multiple interrelated factors. Academic stress continues to dominate as the most significant contributor, particularly during high-stakes assessments or competitive academic settings. Performance expectations from family and institutions further exacerbate pressure. The consistent observation that students suffer from test anxiety, research-related anxiety or subject-specific stress points to a need for academic reforms and emotional skill-building. Additionally, the literature suggests that anxiety is not merely an isolated condition but is often part of a larger web of psychological and emotional difficulties. Many students simultaneously face depression, emotional dysregulation and poor sleep, which compound the effects of anxiety and hinder academic performance and personal well-being.

A gender-based pattern is evident in the data, with female students generally reporting higher anxiety levels across several domains, including academic and social settings. This disparity may stem from biological, cultural or social expectations that place different emotional burdens on female students. First-year students and those in demanding disciplines, such as medical and physiotherapy programmes, also exhibited increased anxiety, pointing to the vulnerability of students during transitional and high-pressure academic periods.

Coping strategies adopted by students were diverse, yet their effectiveness varied. Techniques such as yoga, mindfulness and physical activity showed positive results in reducing anxiety. Structured interventions such as biofeedback training and peer support networks also demonstrated benefits. However, the review notes a significant concern around stigma and low help-seeking behaviour, which limits students' access to available support systems. The onset of the COVID-19 pandemic further intensified anxiety across the student population, with isolation, online learning fatigue and health anxieties aggravating already existing mental health issues. These findings collectively call for proactive and multidimensional approaches to understanding and addressing student anxiety in educational institutions.

This systematic review synthesises evidence from 40 studies to demonstrate that anxiety among undergraduate students is a multifaceted issue, driven by academic pressures, social challenges and psychological comorbidities, with notable disparities across gender, academic stage and regional contexts. The COVID-19 pandemic further exacerbated these trends,

underscoring the urgent need for institutional and policy-level interventions.

Conclusion

This systematic literature review concludes that anxiety is a major and growing concern among undergraduate students across global and regional educational settings. Academic demands, social pressures, emotional vulnerabilities and contextual factors such as pandemics all contribute to the development and intensification of anxiety. The prevalence rates found in the reviewed studies underscore the urgency of this mental health issue, with some reporting anxiety levels exceeding 80% among student populations. The effects of anxiety extend beyond emotional distress, influencing academic performance, social functioning and long-term well-being.

While students do engage in various coping strategies, access to formal mental health support remains inconsistent. Gender, discipline of study and stage of education further moderate the experience and expression of anxiety.²¹ Institutions must therefore recognise that anxiety is not a one-size-fits-all issue and that tailored interventions are required. Holistic and inclusive approaches that integrate mental health awareness, emotional skill development and flexible academic support are essential for promoting student well-being.²² Future research must explore not only causes and correlations but also evaluate long-term interventions across diverse populations to develop more resilient student communities.

A proposed framework of intervention categories and strategies is outlined in Table 13.

Further Recommendations

- **Integrate mental health education** into the undergraduate curriculum, including stress management, emotional regulation and mindfulness practices.
- **Establish or strengthen campus-based counselling centres** to provide accessible, affordable and stigma-free mental health support.
- **Conduct regular mental health screenings** or surveys to identify students at risk and intervene early.
- **Develop peer mentoring and support groups**, especially for first-year students, to ease academic and social transitions.
- **Incorporate relaxation techniques** such as yoga, breathing exercises and guided meditation into co-curricular or wellness activities.
- **Promote help-seeking behaviour** through awareness campaigns and sensitisation programmes that address stigma and normalise mental health care.
- **Provide faculty and staff training** on identifying signs of anxiety and referring students to appropriate services.

Table 13. Intervention Category Related to Specific Strategy.

S. No.	Intervention Category	Specific Strategy	Implementation Example
1.	Targeted interventions for high-risk groups	Gender-specific programmes	Address female students' higher anxiety risk with assertiveness training and safe spaces for social support.
		Transition support for freshmen	Implement mentorship programmes pairing seniors with first-years to ease academic and social adjustment.
2.	Institutional support systems	Scalable counselling services	Expand campus mental health centres with tiered support (peer counselling for mild cases and clinical referrals for severe anxiety).
		Anonymous screening tools	Deploy AI-driven chatbots (e.g., Woebot) for early anxiety detection and triage, linked to university health systems.
3.	Policy reforms	Flexible assessment policies	Replace high-stakes exams with iterative low-stakes evaluations to mitigate performance anxiety.
		Faculty training	Train educators to recognise anxiety symptoms and refer students to resources via workshops on 'Mental Health First Aid'.
4.	Curriculum-integrated mental health education	Mandatory well-being modules	Embed stress management, mindfulness and emotional regulation training into first-year orientation programmes and core curricula.
		Credit-bearing courses	Offer electives on mental health literacy (e.g., 'Psychology of Resilience') to reduce stigma and promote help-seeking.
5.	Research and innovation	Longitudinal tracking	Universities should collaborate with researchers to monitor anxiety trends post-intervention using standardised tools.
		Digital therapeutics	Invest in VR-based exposure therapy for social anxiety and app-delivered CBT.

- **Review academic workload and assessment policies** to reduce unnecessary pressure, such as by implementing flexible deadlines and formative assessments.
- **Target interventions toward high-risk groups**, including female students, medical/technical students and those from disadvantaged backgrounds.
- **Encourage further research** on digital mental health interventions (e.g., mobile apps and online counselling) and their impact on student anxiety.
- **Support longitudinal and comparative studies** to explore how anxiety trends evolve over time and across institutions or countries.

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Key elements this covers:

1. Transparency about AI use
2. Scope of AI assistance (non-substantive vs. substantive)
3. Your oversight as author
4. Conformance with academic integrity standards

Authors' Contribution

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Statement of Ethics

This study is a systematic literature review and did not involve direct interaction with human participants; therefore, ethical approval was not required.

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